



DPULookUpTable

One dimensional lookup table with linear interpolation between sampling points.

1. Inputs:

In Input value for which a value should be retrieved from the lookup table

2. Dynamic parameters:

SamplingPoint[1...100] Sampling points of the lookup table. The first sampling point is *SamplingPoint1*. Default: all = 0.

Sample[1...100] Values corresponding to the sampling points.
Default: all = 0

3. Static parameters:

NSamplingPoints Total number of sampling points (max. 100). Default: 100

4. Outputs:

Out Value corresponding to input value *In* (interpolated)

5. Functional principle:

The sampling points and the corresponding values are defined in the DPU configuration.

6. Exampe:

```
DPULookUpTable lut
{
    Computer = {...};
    Index = ...;

    NSamplingPoints = 3;
    SamplingPoint1 = 0;   Sample1 = -5.3;
    SamplingPoint2 = 0.3; Sample2 = 3.6;
    SamplingPoint3 = 10;  Sample3 = 7.4;
};
```

If *In* lies between two sampling points then *Out* is linearly interpolated between the corresponding samples. If *In* $\leq$ *SamplingPoint1* then *Out* = *Sample1*. If *In* $\geq$ *SamplingPoint[NSamplingPoints]* then *Out* = *Sample[NSamplingPoints]*.